Effect of lifestyle intervention on the reproductive endocrine profile in women with polycystic ovarian syndrome: a systematic review and meta-analysis

Polycystic ovarian syndrome (PCOS) affects 18–22% of women at reproductive age. We conducted a systematic review and meta-analysis evaluating the expected benefits of lifestyle (exercise plus diet) interventions on the reproductive endocrine profile in women with PCOS. Potential studies were identified by systematically searching PubMed, CINAHL and the Cochrane Controlled Trials Registry (1966–April 30, 2013) systematically using key concepts of PCOS. Significant improvements were seen in women receiving lifestyle intervention vs usual care in follicle-stimulating hormone (FSH) levels, mean difference (MD) 0.39 IU/l (95% CI 0.09 to 0.70, P=0.01), sex hormone-binding globulin (SHBG) levels, MD 2.37 nmol/l (95% CI 1.27 to 3.47, P<0.0001), total testosterone levels, MD −0.13 nmol/l (95% CI −0.22 to −0.03, P=0.008), androstenedione levels, MD −0.09 ng/dl (95% CI −0.15 to −0.03, P=0.005), free androgen index (FAI) levels, MD −1.64 (95% CI −2.94 to −0.35, P=0.01) and Ferriman–Gallwey (FG) score, MD −1.01 (95% CI −1.54 to −0.48, P=0.0002). Significant improvements were also observed in women who received exercise-alone intervention vs usual care in FSH levels, MD 0.42 IU/l (95% CI 0.11 to 0.73, P=0.009), SHBG levels, MD 3.42 nmol/l (95% CI 0.11 to 6.73, P=0.04), total testosterone levels, MD −0.16 nmol/l (95% CI −0.29 to −0.04, P=0.01), androstenedione levels, MD −0.09 ng/dl (95% CI −0.16 to −0.03, P=0.004) and FG score, MD −1.13 (95% CI −1.88 to −0.38, P=0.003). Our analyses suggest that lifestyle (diet and exercise) intervention improves levels of FSH, SHBG, total testosterone, androstenedione and FAI, and FG score in women with PCOS.